

**General information**

Wellbore name	7220/8-1
Type	EXPLORATION
Purpose	WILDCAT
Status	P&A
Press release	link to press release
Factmaps in new window	link to map
Main area	BARENTS SEA
Field	JOHAN CASTBERG
Discovery	7220/8-1 Johan Castberg
Well name	7220/8-1
Seismic location	3D survey WG08 -inline 1530 & crossline 3470
Drilling operator	Statoil Petroleum AS
Drill permit	1327-L
Production licence at wellhead	532
Prod. licence for drilling target	532
Drilling facility	POLAR PIONEER
Drilling days	65
Entered date	27.02.2011
Completed date	02.05.2011
Release date	02.05.2013
Publication date	02.05.2013
Purpose – planned	WILDCAT
Reentry	NO
Content	OIL/GAS
Discovery wellbore	YES
1st level with HC, age	MIDDLE JURASSIC
1st level with HC, formation	STØ FM
2nd level with HC, age	EARLY JURASSIC
2nd level with HC, formation	NORDMELA FM
Kelly bushing elevation [m]	23.0
Water depth [m]	374.0
Total depth (MD) [m RKB]	2222.0
Final vertical depth (TVD) [m RKB]	2221.0
Maximum inclination [°]	3.2
Oldest penetrated age	LATE TRIASSIC
Oldest penetrated formation	SNADD FM
Geodetic datum	ED50



NS degrees	72° 29' 28.92" N
EW degrees	20° 20' 2.25" E
NS UTM [m]	8051910.71
EW UTM [m]	678908.52
UTM zone	33
NPDID wellbore	6484

Wellbore history

**General**

Well 7220/8-1 was drilled just west of the Polheim Sub-platform and Loppa High in the Barents Sea. The main objective was to prove an economical volume of hydrocarbons and to establish hydrocarbon contacts in the Stø and Nordmela formations in the Skrugard Prospect. The targeted Skrugard fault block is one of several rotated fault blocks in the licence and a part of the Bjørnøyrenna Fault Complex.

Operations and results

A shallow 8 1/2" pre-spud well (7220/8-U-1) was drilled and logged to 955 m to check for shallow gas due to seismic anomalies around the well location. No shallow gas was encountered. After that wildcat well 7220/8-1 was spudded with the semi-submersible installation Polar Pioneer on 27 February 2011 and drilled to TD at 2222 m in the Late Triassic Snadd Formation. Drilling operations proceeded without significant problems but upon intermediate MDT logging the cable broke and about 10 days were lost fishing. The well was drilled with spud mud and hi-vis pills down to 850 m and with KCl/Polymer/Glycol from 850 m to TD.

The well penetrated Tertiary and Cretaceous Claystones and Sandstones and upper Jurassic Claystones above the reservoir. In the reservoir the well penetrated Sandstones of Jurassic age, within the Stø, Nordmela and Tubåen Formations, and of Triassic age within the Fruholmen and Snadd Formations. The primary target reservoir was penetrated with top Stø Formation at 1276 m, and top Nordmela Formation at 1354 m. These Formations contained a 37 m thick gas column (GOC at 1312 m) and an 83 m thick oil column (OWC at 1395 m). Results from the well indicate that flat spots seen on the geo-seismic section represent the gas-oil-contact (GOC) and the oil-water-contact (OWC). The Snadd Formation was penetrated at 2122 m TVD, which was 35 m shallower than the prognosis.

Very good hydrocarbon shows were seen when drilling the reservoir in the Stø and Nordmela Formations. No shows were observed below 1400 m in the core chips and cuttings.

A total of five cores were cut in the Stø and Nordmela Formations, in the interval 1292.5 - 1405.5 m (97.8% recovery). Schlumberger MDT single probe was used for all samples. Oil samples were acquired at 1320.6 m, 1336.8 m, and at 1380.5 m in the Stø and Nordmela Formations. Due to the broken cable incident no water or gas samples were acquired.

The well was permanently abandoned on 2 May 2011 as an oil and gas discovery.

Testing

No drill stem test was performed.

Cuttings at the Norwegian Offshore Directorate

Cutting sample, top depth [m]	Cutting samples, bottom depth [m]
850.00	2222.00



Cuttings available for sampling?	YES
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Cores at the Norwegian Offshore Directorate

Core sample number	Core sample – top depth	Core sample – bottom depth	Core sample depth – uom
1	1292.5	1296.5	[m]
2	1297.5	1324.2	[m]
3	1324.5	1351.4	[m]
4	1351.5	1377.9	[m]
5	1378.5	1405.0	[m]

Total core sample length [m]	110.5
Cores available for sampling?	YES

Palynological slides at the Norwegian Offshore Directorate

Sample depth	Depth unit	Sample type	Laboratory
850.0	[m]	DC	STATOIL
860.0	[unknown]	DC	STATOIL
870.0	[unknown]	DC	STATOIL
880.0	[unknown]	DC	STATOIL
890.0	[unknown]	DC	STATOIL
900.0	[unknown]	DC	STATOIL
910.0	[unknown]	DC	STATOIL
920.0	[unknown]	DC	STATOIL
930.0	[unknown]	DC	STATOIL
940.0	[unknown]	DC	STATOIL
950.0	[unknown]	DC	STATOIL
960.0	[unknown]	DC	STATOIL
970.0	[unknown]	DC	STATOIL
980.0	[unknown]	DC	STATOIL
990.0	[unknown]	DC	STATOIL
1000.0	[unknown]	DC	STATOIL
1010.0	[unknown]	DC	STATOIL
1020.0	[unknown]	DC	STATOIL
1030.0	[unknown]	DC	STATOIL
1040.0	[unknown]	DC	STATOIL
1050.0	[unknown]	DC	STATOIL
1060.0	[unknown]	DC	STATOIL



1080.0	[unknown]	DC	STATOIL
1090.0	[unknown]	DC	STATOIL
1100.0	[unknown]	DC	STATOIL
1110.0	[unknown]	DC	STATOIL
1120.0	[unknown]	DC	STATOIL
1130.0	[unknown]	DC	STATOIL
1140.0	[unknown]	DC	STATOIL
1150.0	[unknown]	DC	STATOIL
1160.0	[unknown]	DC	STATOIL
1170.0	[unknown]	DC	STATOIL
1180.0	[unknown]	DC	STATOIL
1190.0	[unknown]	DC	STATOIL
1200.0	[unknown]	DC	STATOIL
1203.0	[unknown]	DC	STATOIL
1209.0	[unknown]	DC	STATOIL
1215.0	[unknown]	DC	STATOIL
1221.0	[unknown]	DC	STATOIL
1227.0	[unknown]	DC	STATOIL
1233.0	[unknown]	DC	STATOIL
1239.0	[unknown]	DC	STATOIL
1245.0	[unknown]	DC	STATOIL
1252.0	[unknown]	DC	STATOIL
1258.0	[unknown]	DC	STATOIL
1261.0	[unknown]	DC	STATOIL
1270.0	[unknown]	DC	STATOIL
1276.0	[unknown]	DC	STATOIL
1279.0	[unknown]	DC	STATOIL
1285.0	[unknown]	DC	STATOIL
1288.0	[unknown]	DC	STATOIL
1291.0	[unknown]	DC	STATOIL
1291.0	[unknown]	DC	STATOIL
1292.5	[unknown]	C	STATOIL
1296.5	[unknown]	C	STATOIL
1302.0	[unknown]	C	STATOIL
1307.0	[unknown]	C	STATOIL
1313.0	[unknown]	C	STATOIL
1316.0	[unknown]	C	STATOIL
1324.2	[unknown]	C	STATOIL
1330.0	[unknown]	C	STATOIL
1335.8	[unknown]	C	STATOIL



1340.0	[unknown]	C	STATOIL
1343.8	[unknown]	C	STATOIL
1350.0	[unknown]	C	STATOIL
1351.3	[unknown]	C	STATOIL
1354.0	[unknown]	C	STATOIL
1357.0	[unknown]	C	STATOIL
1358.0	[unknown]	C	STATOIL
1364.0	[unknown]	C	STATOIL
1367.0	[unknown]	C	STATOIL
1371.9	[unknown]	C	STATOIL
1377.0	[unknown]	C	STATOIL
1377.9	[unknown]	C	STATOIL
1382.4	[unknown]	C	STATOIL
1387.5	[unknown]	C	STATOIL
1390.0	[unknown]	C	STATOIL
1396.5	[unknown]	C	STATOIL
1400.0	[unknown]	C	STATOIL
1405.5	[unknown]	C	STATOIL
1414.0	[unknown]	C	STATOIL
1420.0	[unknown]	DC	STATOIL
1426.0	[unknown]	DC	STATOIL
1432.0	[unknown]	DC	STATOIL
1438.0	[unknown]	DC	STATOIL
1444.0	[unknown]	DC	STATOIL
1450.0	[unknown]	DC	STATOIL
1456.0	[unknown]	DC	STATOIL
1462.0	[unknown]	DC	STATOIL
1471.0	[unknown]	DC	STATOIL
1480.0	[unknown]	DC	STATOIL
1489.0	[unknown]	DC	STATOIL
1495.0	[unknown]	DC	STATOIL
1504.0	[unknown]	DC	STATOIL
1510.0	[unknown]	DC	STATOIL
1519.0	[unknown]	DC	STATOIL
1525.0	[unknown]	DC	STATOIL
1531.0	[unknown]	DC	STATOIL
1537.0	[unknown]	DC	STATOIL
1543.0	[unknown]	DC	STATOIL
1549.0	[unknown]	DC	STATOIL
1555.0	[unknown]	DC	STATOIL



1561.0	[unknown]	DC	STATOIL
1567.0	[unknown]	DC	STATOIL
1573.0	[unknown]	DC	STATOIL
1579.0	[unknown]	DC	STATOIL
1585.0	[unknown]	DC	STATOIL
1591.0	[unknown]	DC	STATOIL
1597.0	[unknown]	DC	STATOIL
1603.0	[unknown]	DC	STATOIL
1609.0	[unknown]	DC	STATOIL
1615.0	[unknown]	DC	STATOIL
1621.0	[unknown]	DC	STATOIL
1630.0	[unknown]	DC	STATOIL
1636.0	[unknown]	DC	STATOIL
1642.0	[unknown]	DC	STATOIL
1642.0	[unknown]	DC	STATOIL
1648.0	[unknown]	DC	STATOIL
1654.0	[unknown]	DC	STATOIL
1660.0	[unknown]	DC	STATOIL
1666.0	[unknown]	DC	STATOIL
1672.0	[unknown]	DC	STATOIL
1678.0	[unknown]	DC	STATOIL
1684.0	[unknown]	DC	STATOIL
1690.0	[unknown]	DC	STATOIL
1699.0	[unknown]	DC	STATOIL
1705.0	[unknown]	DC	STATOIL
1711.0	[unknown]	DC	STATOIL
1720.0	[unknown]	DC	STATOIL
1726.0	[unknown]	DC	STATOIL
1738.0	[unknown]	DC	STATOIL
1744.0	[unknown]	DC	STATOIL
1750.0	[unknown]	DC	STATOIL
1756.0	[unknown]	DC	STATOIL
1768.0	[unknown]	DC	STATOIL
1780.0	[unknown]	DC	STATOIL
1789.0	[unknown]	DC	STATOIL
1798.0	[unknown]	DC	STATOIL
1807.0	[unknown]	DC	STATOIL
1816.0	[unknown]	DC	STATOIL
1828.0	[unknown]	DC	STATOIL
1840.0	[unknown]	DC	STATOIL



1849.0	[unknown]	DC	STATOIL
1858.0	[unknown]	DC	STATOIL
1867.0	[unknown]	DC	STATOIL
1876.0	[unknown]	DC	STATOIL
1888.0	[unknown]	DC	STATOIL
1897.0	[unknown]	DC	STATOIL
1906.0	[unknown]	DC	STATOIL
1915.0	[unknown]	DC	STATOIL
1921.0	[unknown]	DC	STATOIL
1930.0	[unknown]	DC	STATOIL
1939.0	[unknown]	DC	STATOIL
1948.0	[unknown]	DC	STATOIL
1954.0	[unknown]	DC	STATOIL
1963.0	[unknown]	DC	STATOIL
1972.0	[unknown]	DC	STATOIL
1978.0	[unknown]	DC	STATOIL
1990.0	[unknown]	DC	STATOIL
1999.0	[unknown]	DC	STATOIL
2008.0	[unknown]	DC	STATOIL
2017.0	[unknown]	DC	STATOIL
2026.0	[unknown]	DC	STATOIL
2032.0	[unknown]	DC	STATOIL
2041.0	[unknown]	DC	STATOIL
2050.0	[unknown]	DC	STATOIL
2059.0	[unknown]	DC	STATOIL
2065.0	[unknown]	DC	STATOIL
2071.0	[unknown]	DC	STATOIL
2080.0	[unknown]	DC	STATOIL
2086.0	[unknown]	DC	STATOIL
2095.0	[unknown]	DC	STATOIL
2104.0	[unknown]	DC	STATOIL
2116.0	[unknown]	DC	STATOIL
2125.0	[unknown]	DC	STATOIL
2134.0	[unknown]	DC	STATOIL
2146.0	[unknown]	DC	STATOIL
2155.0	[unknown]	DC	STATOIL
2161.0	[unknown]	DC	STATOIL
2170.0	[unknown]	DC	STATOIL
2179.0	[unknown]	DC	STATOIL
2188.0	[unknown]	DC	STATOIL



2197.0	[unknown]	DC	STATOIL
2203.0	[unknown]	DC	STATOIL
2212.0	[unknown]	DC	STATOIL
2218.0	[unknown]	DC	STATOIL
2222.0	[unknown]	DC	STATOIL

Oil samples at the Norwegian Offshore Directorate

Test type	Bottle number	Top depth MD [m]	Bottom depth MD [m]	Fluid type	Test time	Samples available
MDT		1380.50	0.00	OIL	13.04.2011 - 00:00	YES

Lithostratigraphy

Top depth [mMD RKB]	Lithostrat. unit
397	NORDLAND GP
455	SOTBAKKEN GP
455	TORSK FM
1014	ADVENTDALEN GP
1014	KOLMULE FM
1227	KOLJE FM
1245	KNURR FM
1252	FUGLEN FM
1276	KAPP TOSCANA GP
1276	STØ FM
1354	NORDMELA FM
1511	TUBÅEN FM
1628	FRUHOLMEN FM
2122	SNADD FM

Geochemical information

Document name	Document format	Document size [MB]
6484_2	.pdf	1.73
6484_1	.pdf	0.14



Logs

Log type	Log top depth [m]	Log bottom depth [m]
CBL CAL	397	865
CMR ECS	839	1436
FMI MSIP	1200	1436
MDT	1251	1472
MDT	1278	1300
MDT	1278	1472
MSIP ZAIT CA CBL	1	100
MSIP ZAIT CA CBL	680	1245
MWD - ARCVRES8 TELESCOPE	852	942
MWD - PERI15 TELE VADN6	942	1472
MWD - TELESCOPE	397	852
PEX HNGS GR	1225	1436
PEX HRLA	1250	2122
USIT	675	1255
VSP	500	2122
VSP TOOL STRING	1000	1255
XPT	1424	2151
ZAIT MSIP	1250	2122

Casing and leak-off tests

Casing type	Casing diam. [inch]	Casing depth [m]	Hole diam. [inch]	Hole depth [m]	LOT/FIT mud eqv. [g/cm3]	Formation test type
CONDUCTOR	30	470.0	36	474.0	0.00	LOT
SURF.COND.	9 5/8	838.0	12 1/4	850.0	1.25	LOT
LINER	7	1251.0	8 1/2	1252.0	0.00	LOT
OPEN HOLE		2222.0	6	2222.0	0.00	LOT

Drilling mud

Depth MD [m]	Mud weight [g/cm3]	Visc. [mPa.s]	Yield point [Pa]	Mud type	Date measured
397	1.60	19.0		Kill Fluid-SW/Bentonite	
397	1.05	15.0		Spud Mud	
471	1.05	14.0		Spud Mud	



472	1.60	19.0		Kill Fluid-SW/Bentonite	
474	1.05	12.0		Spud Mud	
474	1.60	19.0		Kill Fluid-SW/Bentonite	
850	1.05	12.0		Spud Mud	
850	1.20	20.0		KCl/Polymer/Glycol	
850	1.25	21.0		KCl/Polymer/Glycol	
850	1.05	12.0		Spud Mud	
872	1.15	12.0		KCl/Polymer/Glycol	
1035	1.12	13.0		KCl/Polymer/Glycol	
1252	1.12	13.0		KCl/Polymer/Glycol	
1252	1.15	12.0		KCl/Polymer/Glycol	
1379	1.24	15.0		KCl/Polymer/Glycol	
1472	1.21	12.0		KCl/Polymer/Glycol	
1472	1.23	13.0		KCl/Polymer/Glycol	
1472	1.24	12.0		KCl/Polymer/Glycol	
2222	1.21	14.0		KCl/Polymer/Glycol	

Pressure plots

The pore pressure data is sourced from well logs if no other source is specified. In some wells where pore pressure logs do not exist, information from Drill stem tests and kicks have been used. The data has been reported to the NPD, and further processed and quality controlled by IHS Markit.

Document name	Document format	Document size [MB]
6484 Formation pressure (Formasjonstrykk)	.pdf	0.28